Geometry, Perimeter & Area — Level 2 (Compound Shapes)

A dedicated harder follow-on to Week 7, for students who've qualified Level 1. Level 1 asks for the perimeter or area of a single rectangle, square, or triangle; Level 2 combines shapes — paths, frames, cut corners, and figures built from several pieces. All questions are solved using only the plain rectangle/square area and perimeter formulas — no new formulas needed, just combining them. 15 questions, 4-option MCQ, untimed.

Q1. A rectangular garden is 20 m long and 15 m wide. A path of uniform width 2 m is built all around the inside edge of the garden (so the path runs just inside the boundary). Find the area of the path. A) 110 sq m B) 118 sq m C) 124 sq m D) 132 sq m

Q2. A rectangular park is 30 m long and 20 m wide. A path of uniform width 2.5 m is built all around the OUTSIDE of the park. Find the area of the path. A) 255 sq m B) 265 sq m C) 275 sq m D) 285 sq m

Q3. A rectangular photo frame has outer dimensions 40 cm by 30 cm. The wooden border is 5 cm wide all the way around. Find the area of the wooden border itself (not the photo it holds). A) 560 sq cm B) 580 sq cm C) 600 sq cm D) 620 sq cm

Q4. An L-shaped room is made by joining two rectangular sections: one 5 m by 4 m, and another 3 m by 2 m. Find the total floor area of the room. A) 22 sq m B) 24 sq m C) 26 sq m D) 28 sq m

Q5. A square of side 3 cm is cut out from one corner of a rectangular sheet of paper measuring 12 cm by 8 cm. Find the area of the paper that remains. A) 81 sq cm B) 84 sq cm C) 87 sq cm D) 90 sq cm

Q6. If both the length and the breadth of a rectangle are doubled, how many times bigger does its perimeter become? A) 1.5 times B) 2 times C) 3 times D) 4 times

Q7. If the side of a square is tripled (made 3 times as long), how many times bigger does its area become? A) 3 times B) 6 times C) 9 times D) 12 times

Q8. Five identical square tiles, each of side 4 cm, are placed in a row, touching edge to edge, to form one long rectangle. Find the perimeter of this rectangle. A) 40 cm B) 44 cm C) 48 cm D) 52 cm

Q9. Six identical rectangular tiles, each 5 cm by 3 cm, are placed together with no gaps or overlaps to form one bigger rectangle. Find the total area of the bigger rectangle. A) 80 sq cm B) 85 sq cm C) 90 sq cm D) 95 sq cm

Q10. A rectangular field is 40 m long and 25 m wide. A square flower bed of side 5 m is built inside the field. Find the area of the field that is NOT covered by the flower bed. A) 950 sq m B) 960 sq m C) 970 sq m D) 975 sq m

Q11. A rectangular cardboard sheet is 10 cm by 6 cm. A 2 cm by 2 cm square notch is cut out of one corner (the cut edges run parallel to the original sides), leaving an L-shaped piece. Find the perimeter of the remaining L-shaped piece. A) 28 cm B) 30 cm C) 32 cm D) 36 cm

Q12. A figure is made by joining 5 identical squares in a plus-sign (+) shape — one square in the centre, with one square attached to each of its four sides. How many lines of symmetry does this figure have? A) 1 B) 2 C) 4 D) 8

Q13. The three angles of a triangle are 50°, 60°, and x°. Find x. A) 60° B) 65° C) 70° D) 75°

Q14. A figure drawn on squared paper is a 6×5 grid of unit squares, with a 2×2 block of unit squares removed from one corner. Find the area of the figure, in square units. A) 22 B) 24 C) 26 D) 28

Q15. Two square tiles, each of area 49 sq cm, are placed side by side (touching along one full edge) to form a rectangle. Find the perimeter of this rectangle. A) 35 cm B) 38 cm C) 42 cm D) 49 cm

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Answer key: `answer-key.md`